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Africa's leap ahead into cloud: Opportunities and barriers

Early indications hint at an impressive surge into the cloud by some African businesses—and the potential to unlock much greater value.

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Despite significant challenges, companies in Africa have shown they are no laggards when it comes to adopting cloud. Early indications, in fact, show that Africa is moving quickly onto cloud, and there are no signs of slowing down.

In a recent McKinsey survey of technology leaders at more than 50 major African businesses, the participants reported having, on average, about 45 percent of their workloads in public cloud today. Although based on a relatively small sample size, that's on par or ahead of the rates of adoption in North America and China.

The history of mobile adoption on the African continent provides an intriguing reference point. The population of sub-Saharan Africa in 2000, for example, was 90 percent less likely to have access to a fixed telephone than people in East Asia or Latin America. But they had urgent and important needs, such as voice connectivity and eventually mobile banking, which led to a rapid adoption of mobile. By 2020, the region had more than half a billion unique mobile subscribers, making Africa the fastest-growing area for mobile technology.[1] Similar dynamics—limited legacy infrastructure and innovative technology advances—are in place when it comes to cloud, providing African businesses with another opportunity to leapfrog ahead.

African companies that can make the leap stand to gain a sizable prize. Recent McKinsey research projected a global cloud value of $3 trillion, with $797 billion of this value sitting in Africa and Europe across what we've categorized as the Rejuvenate dimension (IT cost efficiencies) and the Innovate dimension (revenue uplifts and business operations savings).[2] Many of Africa's businesses are in sectors where cloud can have a significant impact, such as banking, telecommunications, and oil and gas.

Challenges abound, however, in capturing this value. Wide variations in language, culture, and currency exist, and generally lower market maturity levels, differing regulatory environments, limited infrastructure, and a shortage of colocation data centers[3] hamper cloud programs. And despite lower income levels than other countries, cloud service providers (CSPs) in Africa sometimes cost more on average for the same services than they do in other regions around the world.[4]

To better understand the current state of cloud adoption in the region and more precisely identify the use cases and challenges facing African countries, we surveyed more than 50 technology executives across four primary sectors in Africa and conducted detailed interviews with more than a dozen others (see sidebar, "About the research").

[1] "Leapfrogging," Institute for Security Studies (ISS) African Futures, October 27, 2023; and Francois Botha, "Why Africa has the ability to leapfrog the rest of the world with innovation," *Forbes*, April 2, 2019.
[2] "Projecting the global value of cloud: $3 trillion is up for grabs for companies that go beyond adoption," McKinsey, November 28, 2022. The analysis did not differentiate between Africa and Europe when it comes to cloud's value potential. While much of the value potential lies in Europe, a significant amount is available for Africa.
[3] For example, AWS has one region in Africa (Cape Town), Azure has one (Johannesburg), and Google Cloud does not yet have any.
[4] Example using AWS Pricing Calculator for cloud services in Cape Town versus other regions.

About the research

In 2022 and 2023, we surveyed 53 tech executives working in businesses in the following sectors: financial services; consumer packaged goods; telecommunications, media, and technology; and global energy and minerals. We also had detailed discussions with another dozen executives. These leaders spanned most African regions[1] (except Central Africa) and were directly involved in their organizations' key technology decisions, as C-suite members, vice presidents, or IT directors. Company sizes ranged up to (and a few over) $1 billion a year in revenues, with more than half of participants coming from lower-revenue companies (less than $50 million a year).

Executives were asked a variety of questions about their cloud footprint today, migration strategies, advantages and obstacles encountered in cloud adoption, and their future plans for cloud.

The survey and analysis were not intended to create a holistic view of the market but rather to generate indicative insights into cloud trends in Africa.

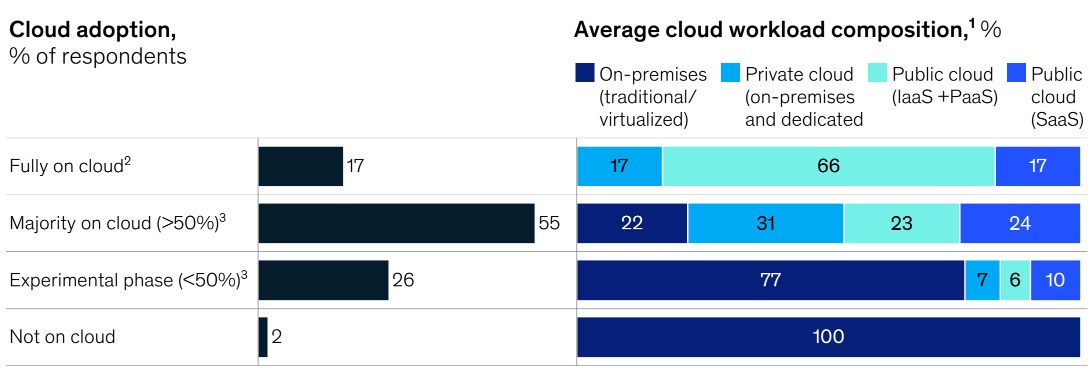
[1] As defined by the United Nations Statistics Division (UNSD).

Cloud adoption in the African market today

Of the African executives we surveyed and spoke to, most have transitioned at least some workloads to a cloud environment. Almost a fifth of these companies have all workloads in some sort of cloud environment, with most workloads in public cloud—infrastructure as a service (IaaS), platform as a service (PaaS), software as a service (SaaS)—and others in private cloud (dedicated and on-premises) (Exhibit 1). Cloud expenditure across organizations interviewed was sizable, accounting for an average of 38 percent of total IT expenditure, compared to projected global cloud spend of about 50 percent of IT expenditure by 2025.[5] This trend was particularly evident among smaller organizations, where the ratio of cloud-to-IT spend was higher on average.

Exhibit 1

Only 17 percent of respondents are fully on cloud.



[1]Breakdown includes compute, storage, network, and database.
[2]Either dedicated private cloud, IaaS, PaaS, or SaaS, with no workloads on-premises (traditional/virtualized).
[3]Includes on-premises (traditional/virtualized) and private cloud (on-premises and dedicated).

[5] "Gartner says more than half of enterprise IT spending in key market segments will switch to the cloud by 2025," Gartner press release,February 9, 2022.

These numbers compare favorably with adoption rates seen in similar surveys in other regions. Recent Cloud by McKinsey research in North America, based on interviews with roughly 30 major companies using cloud, found that the public-cloud adoption rate was on average around 40 percent. In China, a Cloud by McKinsey survey of 278 decision makers found that the average percent of workloads hosted in public cloud was around 30 percent.[6]

Respondents cited key Africa-specific issues—such as critical infrastructure challenges, power needs (load shedding, for example), and local supply chain shortages of on-premises hardware—as drivers of adoption. A limited history of building up legacy systems and incurring technical debt that can complicate migrations is likely another contributing factor.

Digging beneath the surface of the overall adoption rates, we found that only a small percentage of the respondents (26 percent) were in the "experimental phase," with less than 50 percent of workloads in cloud (public or private). The majority have already deployed cloud technology to multiple or all business units. Compute and storage make up the majority of cloud services used (29 percent and 32 percent, respectively).

Leading organizations are now actively using the benefits of cloud technology—scalability, support elasticity, services, and operational efficiencies—to support use cases such as creating end-to-end digital journeys, building new products, and running customer analytics. One African financial institution, for example, recently launched a new digital bank completely on public cloud, which allows it to scale the bank's compute and storage capacity as more customers are onboarded.

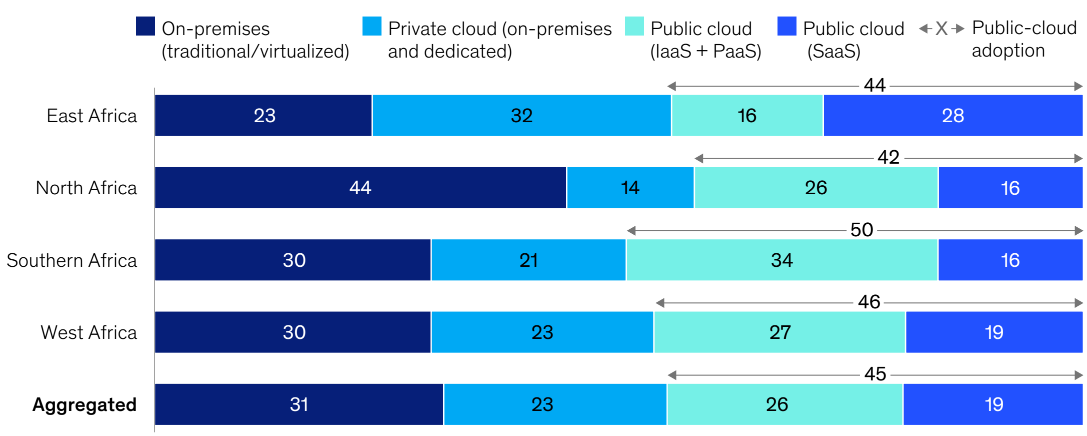
Regional consistency, sector variation

Cloud adoption among respondents is fairly consistent across African regions,[7] with the highest levels, 70 to 77 percent, in East Africa, West Africa, and Southern Africa (Exhibit 2). In terms of public cloud, participants from Southern Africa reported the highest usage of IaaS, PaaS, and SaaS (50 percent), followed by West

Exhibit 2

Respondents in North Africa had a higher percentage of workloads on-premises compared to those in other African regions.

Average cloud workload composition by African region[1]: Current rate, %



[1]As defined by the United Nations Statistics Division (UNSD).

[6] Kai Shen, Anand Swaminathan, Xiaoxiao Tong, and Kevin Wei Wang, "Cloud in China: The outlook for 2025," McKinsey, July 8, 2022.
[7] As defined by United Nations Statistics Division (UNSD). Research was not able to include organizations in Middle Africa.

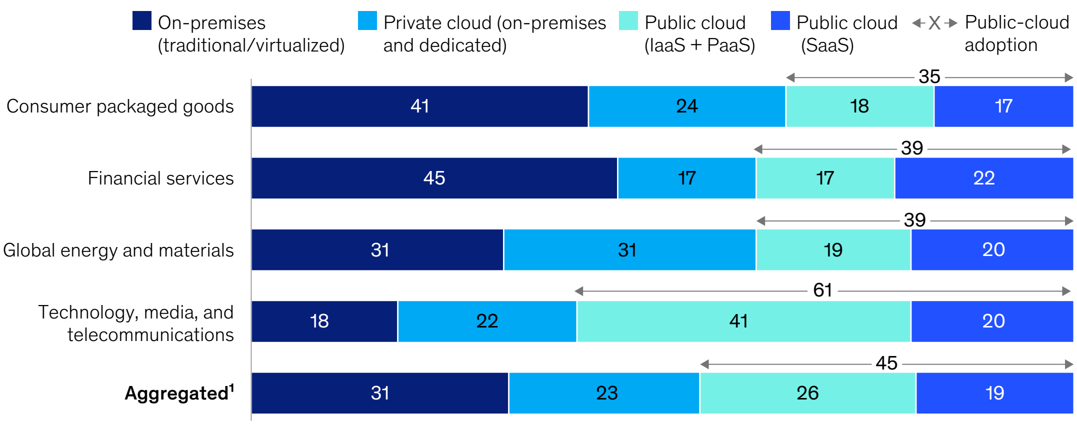
Africa (46 percent) and East Africa (44 percent). Even organizations in other regions with lower cloud presence (such as North Africa), however, reported plans to shift workloads and catch up in the near future.

The findings are indicative of more pronounced differences in adoption rates by sector, where adoption and specific service needs varied depending on the use cases. Participants from technology, media, and telecommunications (TMT) had the highest overall average adoption rate, at 83 percent, primarily in public cloud (Exhibit 3).

Exhibit 3

TMT respondents appear to be leading adoption, with 61 percent of workloads hosted on public cloud.

Average cloud workload composition by industry: Current rate, %



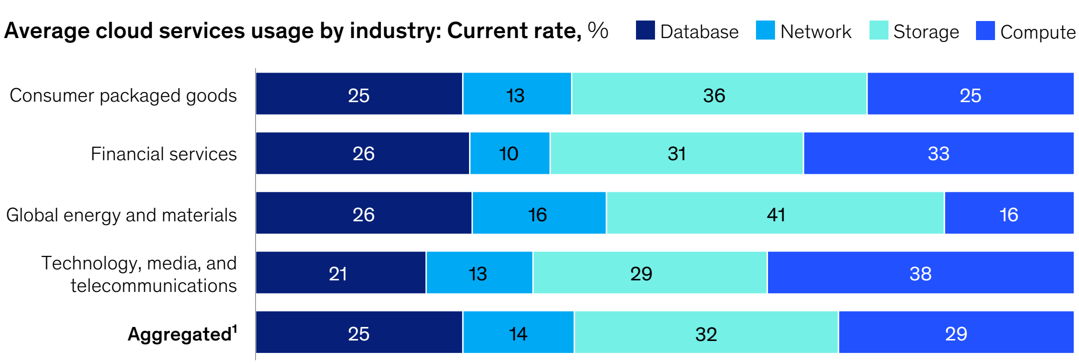
Note: Figures may not sum to 100%, because of rounding.
[1]Includes organizations in research/consulting, nongovernmental organizations, and public sectors.

TMT participants also had the highest proportion of compute services being used in cloud (compared to other services—database, network, storage), suggesting that their heavy cloud adoption is correlated with computationally and analytically intensive use cases (Exhibit 4).

Exhibit 4

Respondents' usage of cloud services varies widely across industries with differing use cases.

Average cloud services usage by industry: Current rate, %



Note: Figures may not sum to 100%, because of rounding.
[1]Includes organizations in research/consulting, nongovernmental organizations, and public sectors.

Financial services organizations had the lowest rates of cloud adoption, at an average of 56 percent of workload (39 percent in public cloud, 17 percent in private cloud), likely due in part to regulatory restrictions. Similarly, the kinds of cloud services used also varied. Although global energy and materials as well as consumer packaged goods respondents were in the middle of the pack in terms of adoption, for example, they both had high rates of using storage services (about 40 percent of their total cloud services usage). Location of data storage is often scrutinized, and industries facing fewer regulations may have more flexibility.

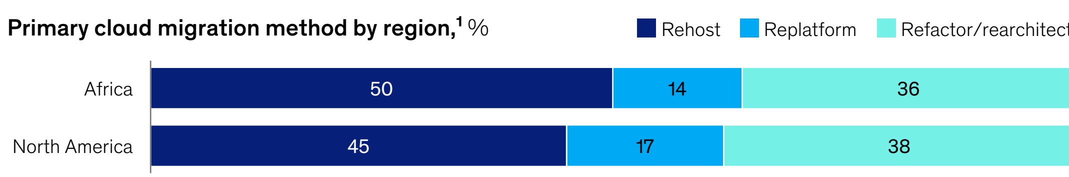
All industries surveyed seemed to find industry-specific use cases that benefited from cloud, and most planned to grow their cloud presence in the near future. One financial services company, for example, has begun using automation and generative AI to reduce the human interaction required to service clients and to provide recommendations to sales staff and instant support help to customers. As its on-premises infrastructure could not keep up with the rapid growth in its compute and data storage needs, it turned to public cloud for these functions. The IT organization at a telecom provider is in the process of migrating 40 percent of its on-premises IT applications to public cloud over the next three years to better manage spikes in demand across geographies.

Cloud adoption approaches

Examining cloud adoption approaches, we found survey participants migrated 50 percent of applications, on average, to rehost workloads, which is a similar rate to interviewed North American companies (Exhibit 5). While rehosting is typically the quickest and cheapest way to migrate an application to cloud, many participants found that this limited their company's ability to leverage many of cloud's benefits.

Exhibit 5

Migration approach of African respondents closely resembles that of North American respondents in similar research effort.



[1]Data for North America (n = 47) sourced from Cloud by McKinsey research effort; comparison excludes repurchase/replace.

In contrast, the most common migration focus (about 40 percent) for surveyed companies in Southern Africa and in the TMT industry is refactoring or rearchitecting applications, which has allowed them to take better advantage of improved scalability, speed, agility, and security offered by cloud, and better modernize their operating model in parallel. Organizations in other regions and industries report plans to begin refactoring or rearchitecting a greater proportion of their migrated applications as they reach higher maturity levels.

African companies we surveyed clearly favored public over private cloud (in contrast, companies in China favor private-cloud options). On average, participants we spoke to had 45 percent of workloads in public cloud (including about 20 percent in SaaS), with another 23 percent in private cloud. The reason for this discrepancy is likely that organizations struggle to stand up private cloud at scale, especially given Africa's infrastructure and supply chain limitations, compared to the quick access and breadth of services offered by public cloud.

There seemed to be no strong trend among African participants in terms of the number of cloud service providers they used. Half of the respondents, for example, used only one CSP (compared to 87 percent of companies in North America and 76 percent in China). This trend is likely due to the smaller presence of CSPs in Africa, as well as regulatory and data sovereignty constraints.[8]

Strong indicators for increased cloud adoption in Africa

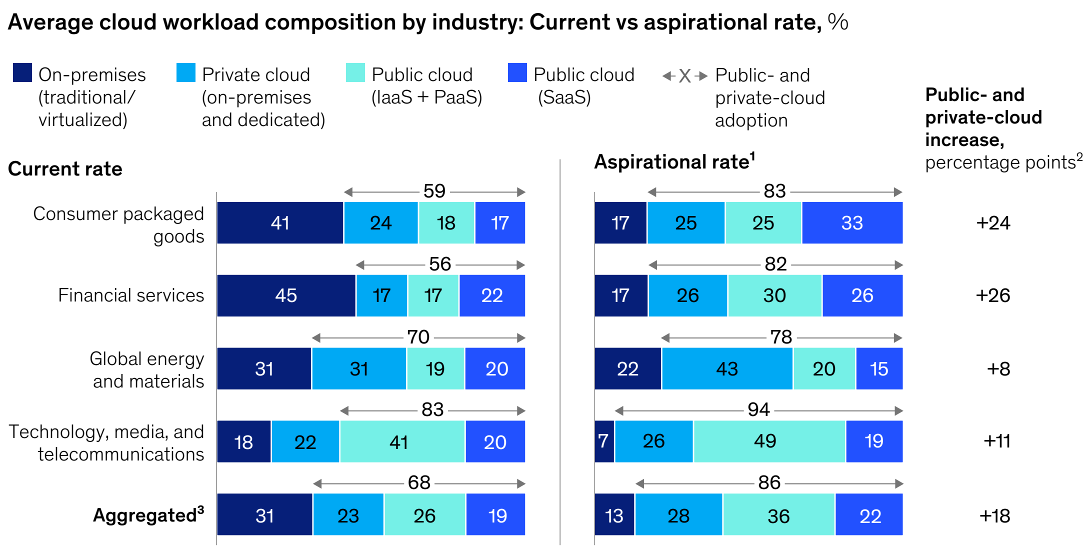
While preferences for future cloud adoption are not necessarily predictive, they are useful indicators of companies' priorities. All respondents felt confident that their organizations would expand their cloud presence in some way over the next one to three years, with participants from financial services predicting the highest rate of increase in public- and private-cloud growth on average, at 26 percentage points (Exhibit 6). However, a few participants, particularly within the global energy and materials industry, predicted more growth in private cloud than public. The majority of respondents (about 60 percent) also believed that most companies in Africa will adopt cloud broadly across their business in the future.

The vast majority of participants said they plan on using cloud for new-application development. Half of them plan on taking a cloud-native approach for all new applications, while the rest cited application criticality, regulatory requirements, latency requirements, and skills availability as dependent factors.

CSPs are noticing this potential. There have been significant investments from global and local cloud providers, for example, in building out African cloud offerings, as well as recent investments in African infrastructure that will support cloud technology, such as fiber-optic cables and broadband infrastructure.[9]

Exhibit 6

All respondents predicted growth in some mix of private- and public-cloud workloads.



Note: Figures may not sum to 100%, because of rounding.
[1]2022 serves as the benchmark for the current rate, while 2025 represents the desired or aspirational rate.
[2]Percentage-point difference between the current rate and aspirational rate of cloud adoption.
[3]Includes organizations in research/consulting, nongovernmental organizations, and public sectors.

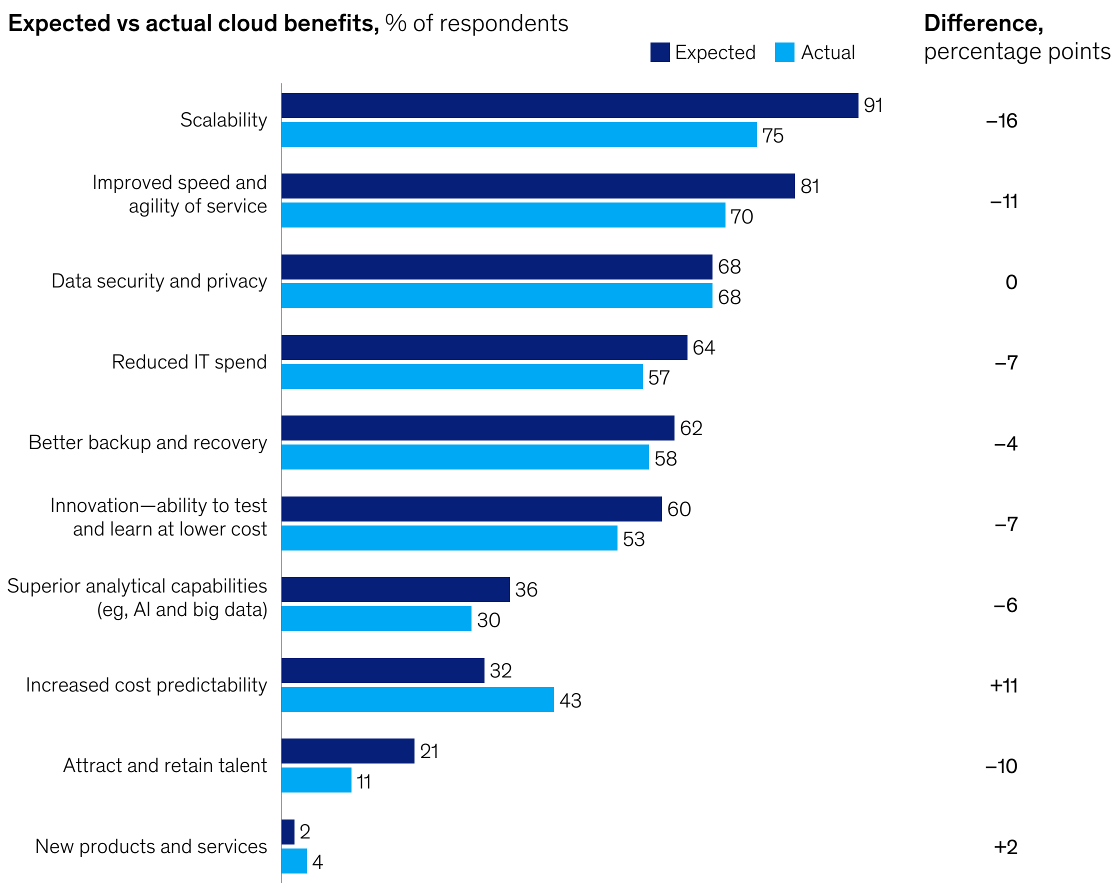
[8] CloudSights database 2022–23, Cloud by McKinsey.
[9] Vaughan O'Grady, "IFC invests to support Seacom's expansion," Developing Telecoms, May 4, 2022.

Challenges faced by African organizations in the cloud

In line with international trends, many organizations in Africa have struggled to realize the full set of benefits they hoped to see from cloud despite progress in increasing cost predictability and adding new products and services (Exhibit 7). The lag in value realization suggests that the cloud adoption journey for African companies has proven to be more complex than anticipated.

Exhibit 7

Majority of expected cloud benefits are not being fully realized.

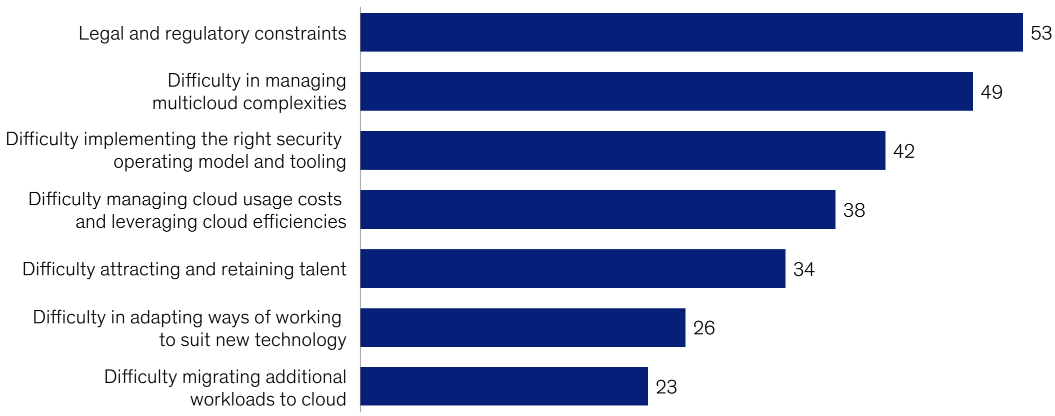


While many of the challenges are the same as those faced by their counterparts in other regions—such as managing costs, security, and operating model—a few of them are more pronounced in Africa. Legal and regulatory constraints were by far the most-cited challenge, with over 50 percent of respondents casting them as significant roadblocks (Exhibit 8). Only 10 percent of respondents said that the legal and policy requirements currently in place support cloud adoption.

Exhibit 8

Legal and regulatory constraints and multicloud complexities are seen as the two biggest barriers to cloud adoption.

Factors impacting current cloud adoption, % of respondents



Data residency laws, for example, such as articles included in Algeria's, Gabon's, Niger's, and Morocco's data protection laws, are common across Africa and require regulated data (such as personal information) to be localized within the country's borders.[10] Given the limited data center presence of cloud providers in Africa, this regulation essentially makes it impossible for many organizations to use public cloud for these data sets.

Other countries in Africa (for example, Kenya, South Africa, Tunisia, and Uganda) also have restrictions against cross-border data transfer.[11] Cross-border data transfer is a critical cloud use case for most multinational companies, with financial services organizations likely to be particularly affected. In the future, regulatory bodies, organizations, and service providers will need to collaborate to establish policies and frameworks that address legal and regulatory concerns while encouraging cloud adoption and innovation rather than prohibiting it.

The ability to attract and retain the required cloud talent is another challenge with unique complexities on the Africa continent. Some 97 percent of African organizations expect to have a tech skills challenge in 2023, with attracting skilled new recruits as their number one complaint.[12] Furthermore, those companies that have had success in attracting talent have trouble keeping them in the face of strong incentives to move to a higher-paying region (or find remote work).

These issues are forcing African organizations to reconsider their sourcing models for highly skilled IT talent and to create better attraction and incentive strategies. One area of focus is establishing strong partnerships with system integrators and CSPs to get support on their cloud migration while simultaneously upskilling existing teams.

[10] "Which way for data localization in Africa?" Collaboration on International ICT Policy for East and Southern Africa (CIPESA), November 2022.
[11] Ibid.
[12] *Africa's tech skills scarcity revealed*, SAP Africa, March 3, 2023.

While the results of our survey indicate that companies in Africa seem to be following global trends toward increased cloud adoption, they still have a significant journey ahead of them. An increased focus on addressing local issues, coupled with a more cloud-friendly environment, are promising signs that companies in Africa are expanding their opportunities to capture a greater share of cloud's significant potential value.

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